

489 translocate at either side of the membrane, and only the fully loaded carrier (EAB) is able to cross
 490 the membrane.

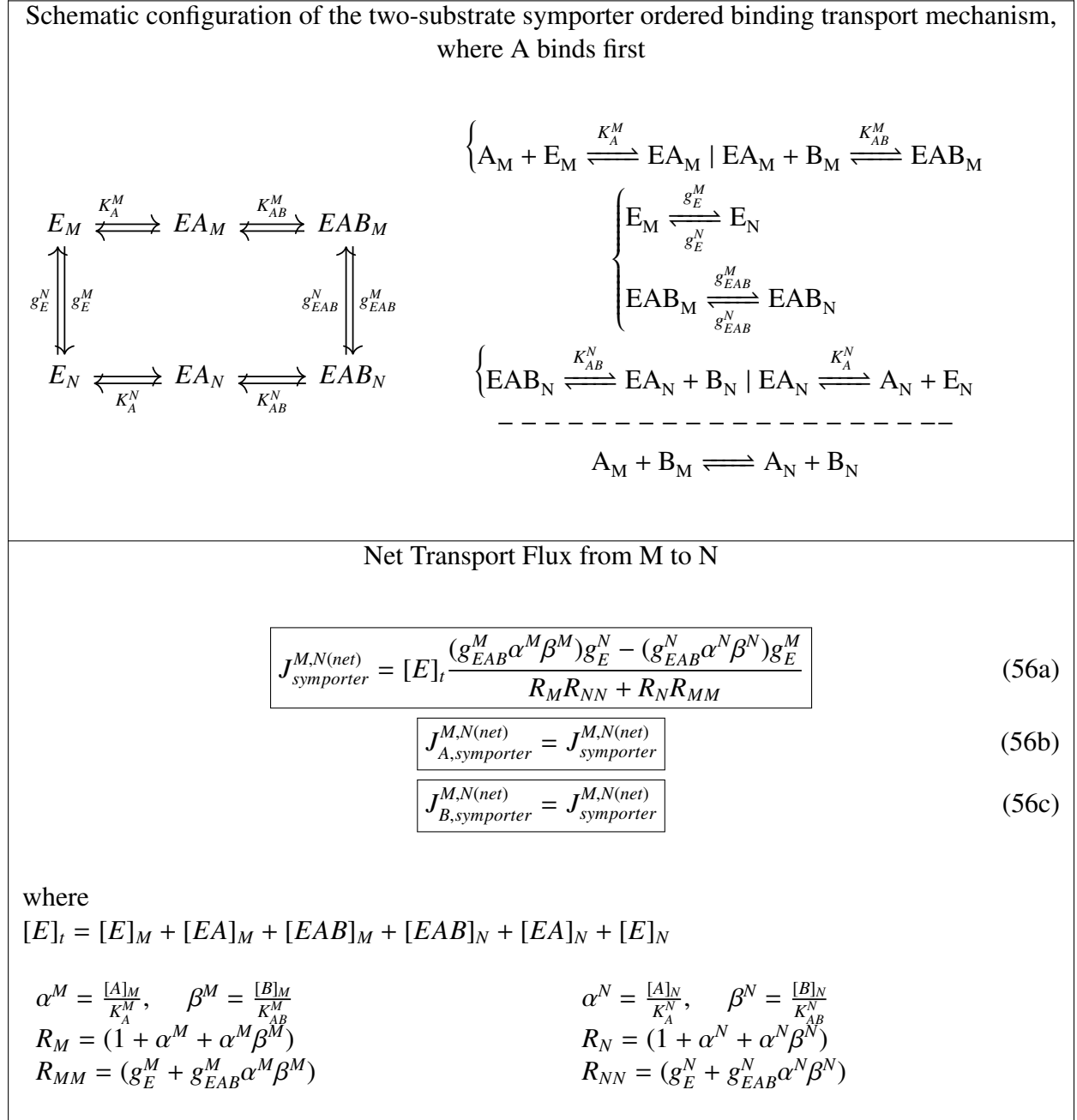
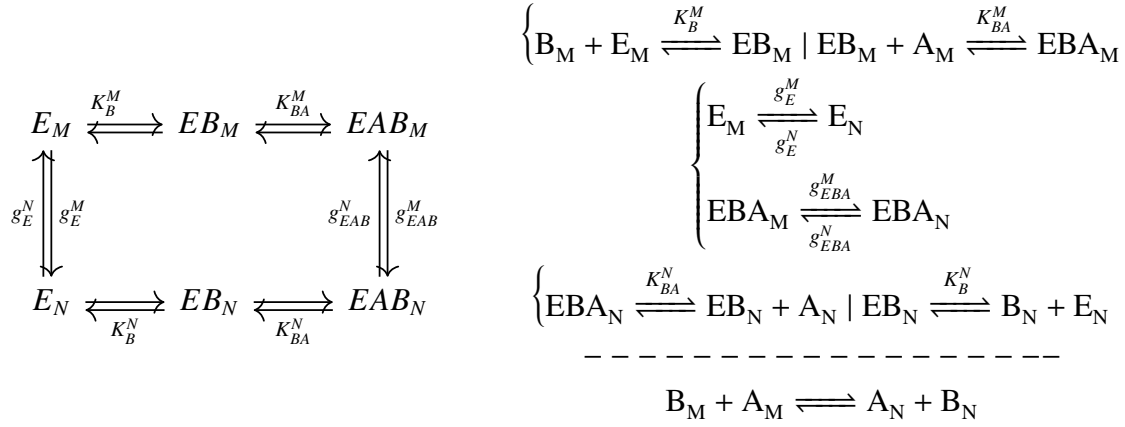


Figure 12: A schematic configuration and the regarding net flux transport equation of the symporter ordered binding with no slippage, for transport of two solutes where A binds first. Dissociation constants (K_A and K_{AB} terms) and the translocation rate constants (g_E and g_{EAB} terms) are depicted in the reaction cycle in the left panel. $[E]_t$ is the total number of transporter per unit area of the membrane. The flux equation is transformed to the general form of the resistance from Stein model [131].

Schematic configuration of the two-substrate symporter ordered binding transport mechanism, where B binds first



Net Transport Flux from M to N

$$J_{symporter}^{M,N(net)} = [E]_t \frac{(g_{EAB}^M \alpha'^M \beta^M) g_E^N - (g_{EAB}^N \alpha'^N \beta^N) g_E^M}{R_M R_{NN} + R_N R_{MM}} \quad (57a)$$

$$J_{A,symporter}^{M,N(net)} = J_{symporter}^{M,N(net)} \quad (57b)$$

$$J_{B,symporter}^{M,N(net)} = J_{symporter}^{M,N(net)} \quad (57c)$$

where

$$[E]_t = [E]_M + [EB]_M + [EAB]_M + [EAB]_N + [EB]_N + [E]_N$$

$$\begin{aligned}
 \alpha'^M &= \frac{[A]_M}{K_{BA}^M}, & \beta^M &= \frac{[B]_M}{K_B^M} \\
 R_M &= (1 + \beta^M + \alpha'^M \beta^M) \\
 R_{MM} &= g_E^M + g_{EAB}^M \alpha'^M \beta^M
 \end{aligned}$$

$$\begin{aligned}
 \alpha'^N &= \frac{[A]_N}{K_{BA}^N}, & \beta^N &= \frac{[B]_N}{K_B^N} \\
 R_N &= (1 + \beta^N + \alpha'^N \beta^N) \\
 R_{NN} &= g_E^N + g_{EAB}^N \alpha'^N \beta^N
 \end{aligned}$$

Figure 13: A schematic configuration and the regarding net flux transport equation of the symporter mechanism with no slippage, for transport of two solutes where binding happens in order and B binds first. Dissociation constants (K_{BA} and K_B terms) and the translocation rate constants (g_E and g_{EAB} terms) are depicted in the reaction cycle in the right panel. $[E]_t$ is the total number of transporter per unit area of the membrane. The flux equation is transformed to the general resistance form from Stein model [131].

Three-substrate symporter ordered binding model:

Figure 14 shows a kinetic scheme for a carrier, E , that is capable of transporting molecules of A ,